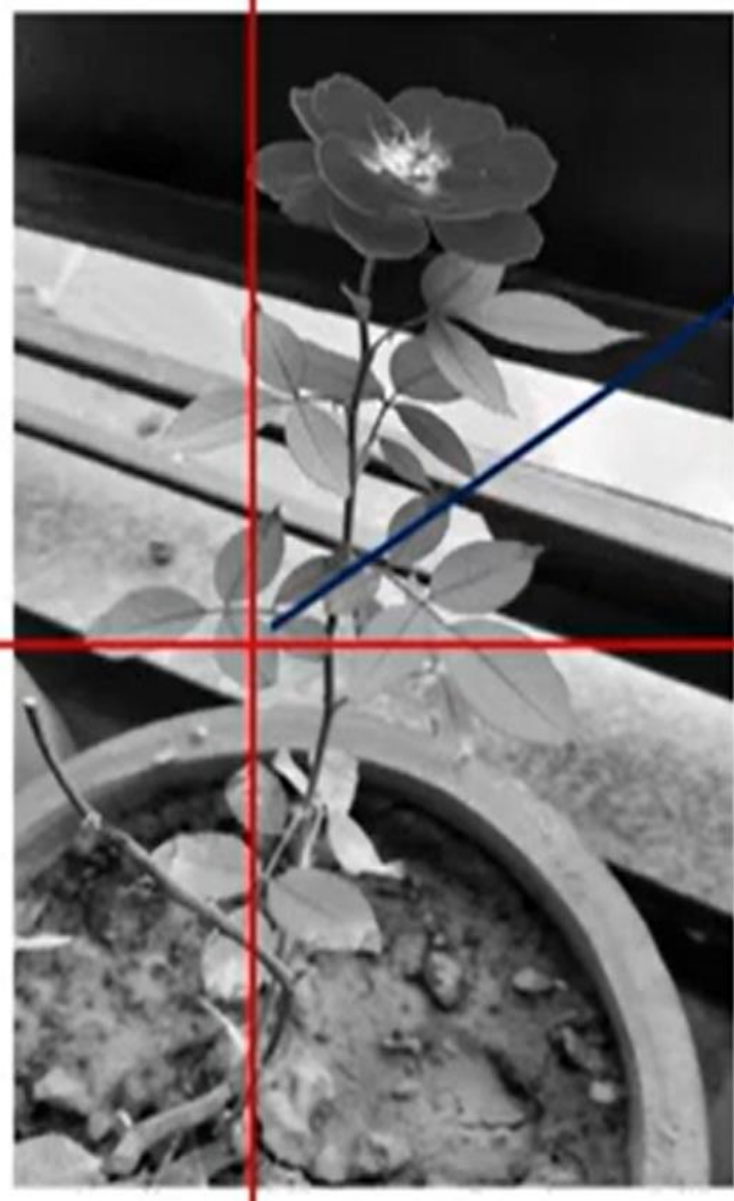


What is an Image?

- Image is an pictorial representation of someone or something.
- Someone has truly said “ A picture is worth a thousand words”.



- An image is a two-dimensional function $f(x,y)$, where x and y are the spatial (plane) coordinates, and the amplitude of f at any pair of coordinates (x,y) is called the intensity of the image at that level.
- If x,y and the amplitude values of f are finite and discrete quantities, we call the image a digital image. A digital image is composed of a finite number of elements called pixels, each of which has a particular location and value.



$f(x_0, y_0)$



$f(x_1, y_1)$

Pixel intensity value

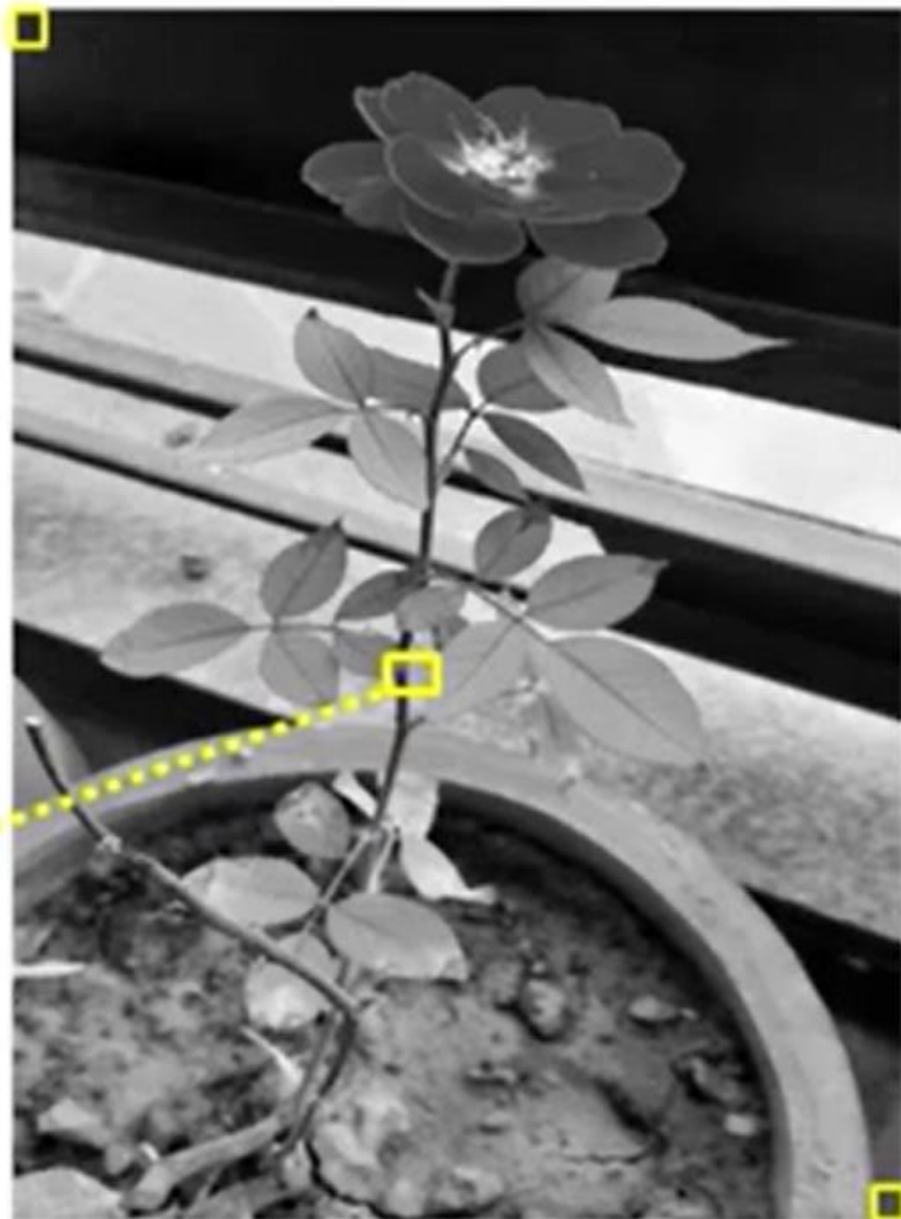
$$f(1,1) = 33$$

Pixel location

rows columns
↑ ↑

$$f(645:650,1323:1328) =$$

83 82 132 132 131 130
82 82 132 132 131 130
82 82 132 132 131 130
82 82 132 132 132 131
80 79 133 133 132 131
80 79 133 133 132 131



$$f(2724,2336) = 83$$

Consider the following image (2724x2336 pixels) to be 2D function or a matrix with rows and columns

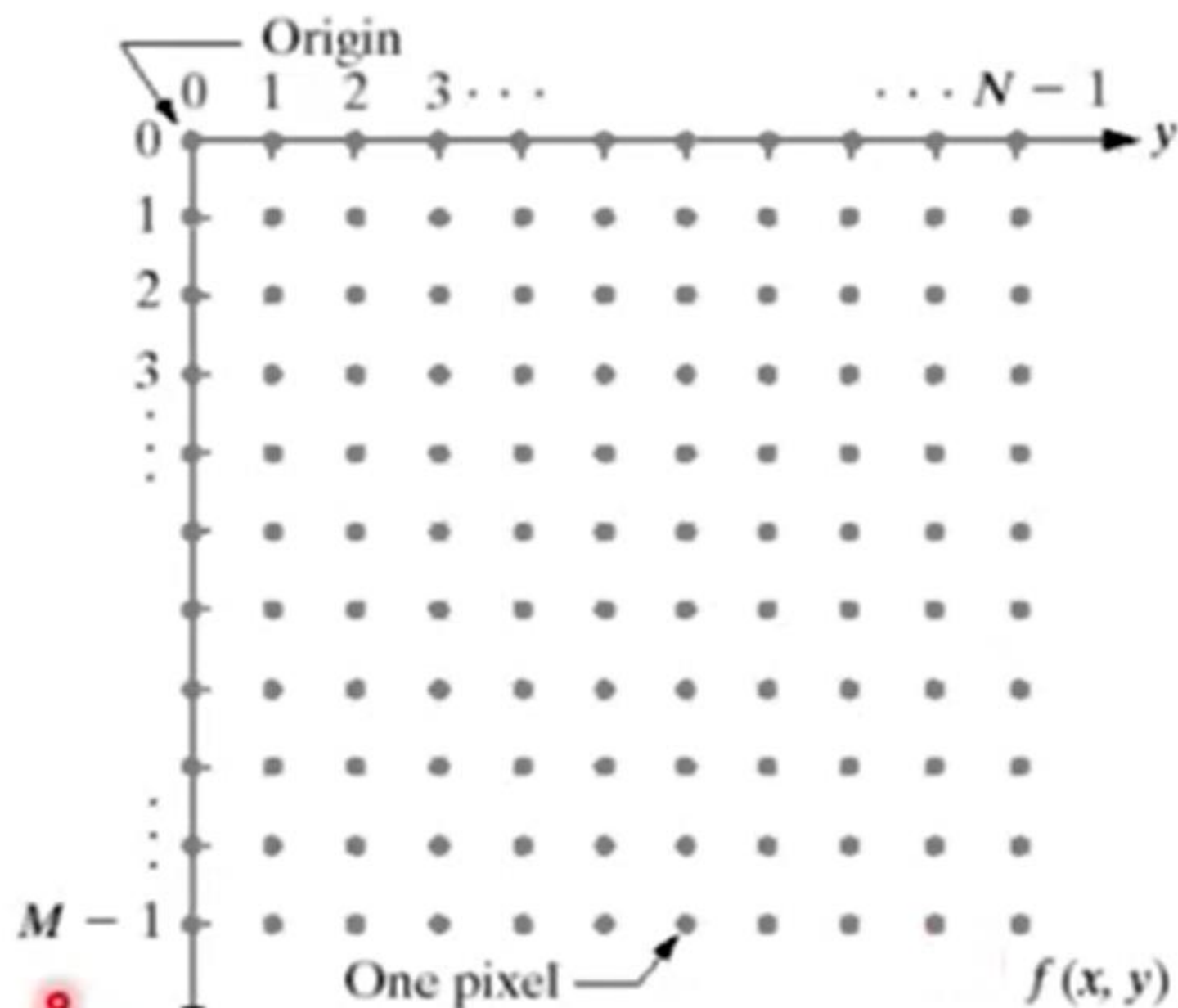
In 8-bit representation
Pixel intensity values
change between 0 (Black)
and 255 (White)



One Pixel



Conventional coordinate for image representation



- Each pixel represent by k-bits, that means image will have 2^k different values.
- For 8 bit representation,

$$2^8 = 256 \text{ different values of pixels}$$

For 1 bit representation,

$$2^1 = 2 \text{ different values of pixels}$$

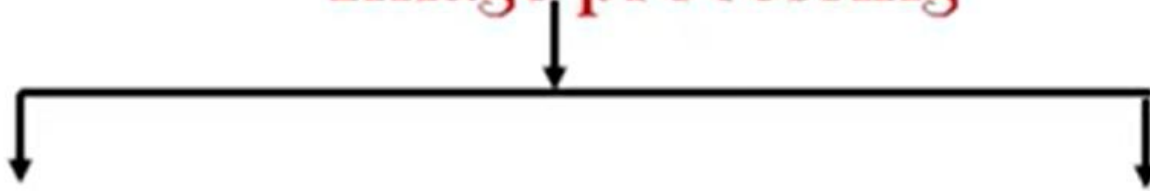
What is Image Processing?

Image processing is manipulation of images by a brain or computers.



Types of Image Processing

Image processing



Analog Image processing

E.g.: Processing CRT TV image



Digital Image processing

E.g.: Processing image using PCs, Smartphones



What is Digital Image Processing?

- Digital Image Processing is a method to perform some operations on an digital image, in order to get an enhanced image or to extract some useful information from it.
- It is a type of signal processing in which input is an image and output may be image or characteristics/features associated with that image.
- The **motivation** or purposes behind DIP is:
 - (1) Improvement of picture information for interpretation.
 - (2) Storage, transmission and representation of digital data for machine perception is possible.

Types of Digital Image Processing

- (1) **Low level processing:** Primitive operations such as noise reduction, image sharpening, enhancement etc. Input and output are images.
- (2) **Mid level processing:** Image segmentation, classification of individual objects etc. Here input are images but output are attributes of images for e.g. edges of image.
- (3) **High level processing:** It involves making sense of recognized objects and performing functions associated with visions. For e.g. Automatic character recognition, military recognition, autonomous navigation etc.

Applications of Digital Image Processing

1. Biometrics
2. Vehicle number plate detection
3. Content based Image retrieval (CBIR)
4. Steganography
5. Medical imaging
6. Object recognition
7. Image enhancement and noise removal.